Discrete-Time Markov Chains — Complete Formula Reference

Every identity from Ch. 4.1, zero derivations, plus the extra results your 20-problem set used.

1. Transition Matrix & Markov Property

- **Definition:** $P_{ij} = P(X_{n+1} = j \mid X_n = i)$, row = "from", column = "to"
- **Row-stochastic:** $P_{ij} \geq 0$, $\sum_j P_{ij} = 1$ for every row
- **Markov property:** $P(X_{n+1} = j \mid X_n = i, \text{ any history}) = P_{ij}$
- **Path probability:**

$$P(X_0{=}i_0, \ldots, X_n{=}i_n) = \mu_0(i_0) \prod_{k=1}^{n} P_{i_{k-1} i_k}$$

- **Distribution evolution:** $\mu_{n+1} = \mu_n P$, so $\mu_n = \mu_0 P^n$ (row vector times matrix)

trap: columns need not sum to 1 (only rows do) — unless the matrix is doubly stochastic (Sec. 5).

2. Multi-Step Transitions P^n

- **n-step probability:** $P^n_{ij} = P(X_{m+n} = j \mid X_m = i)$
- **Chapman–Kolmogorov:** $P^{m+n} = P^m P^n$, i.e.

$$P^{m+n}_{ij} = \sum_k P^m_{ik} P^n_{kj}$$

- **One entry of P^2:** dot product of row i with column j — no need for the full matrix
- **Two-state chain** ($p = P_{12}$, $q = P_{21}$): second eigenvalue $\lambda_2 = 1 - p - q$, and rows of P^n approach π like $|1 - p - q|^n$
- **Mixing speed:** $\|P^n - \Pi\| \sim |\lambda_2|^n$; time to get within $\varepsilon \approx \dfrac{\ln \varepsilon}{\ln |\lambda_2|}$

trap: P^n_{ij} sums over all intermediate paths — it is not $(P_{ij})^n$.

3. State Classification

- **Accessible / communicate:** $i \to j$ if $P^n_{ij} > 0$ for some n; $i \leftrightarrow j$ if both ways. **Irreducible** = a single communicating class
- **Absorbing:** $P_{ii} = 1$
- **First-return probability:** $f_{ii} = P(\text{ever return to } i \mid X_0 = i)$
- **Recurrent:** $f_{ii} = 1 \iff \sum_n P^n_{ii} = \infty$
- **Transient:** $f_{ii} < 1 \iff \sum_n P^n_{ii} < \infty$; expected visits (incl. time 0) $= \dfrac{1}{1 - f_{ii}}$
- **Positive recurrent:** recurrent with finite mean return time $m_i < \infty$
- **Null recurrent:** recurrent but $m_i = \infty$
- **Class property:** recurrence, transience, null/positive and period are *shared* by every state in a communicating class
- **Finite chains:** at least one recurrent class; a finite irreducible chain is positive recurrent; no null recurrence

trap: a state can be recurrent (you will return) yet null (the average return time is infinite) — the symmetric random walk on $\mathbb{Z}$ is exactly this.

4. Absorption & First-Step Analysis

- **Absorption probability** into set A: $h_i = \sum_j P_{ij} h_j$, with $h_i = 1$ on A and 0 on other absorbing states
- **Expected absorption time:** $k_i = 1 + \sum_j P_{ij} k_j$, with $k = 0$ on absorbing states
- **Fair gambler's ruin on $\{0, \ldots, N\}$:** $P(\text{hit } N \mid \text{start } k) = \dfrac{k}{N}$
- **Pattern waiting times:** build suffix-state chain, one first-step equation per state. E.g. $\mathbb{E}[\text{flips to HH}] = 6$, $\mathbb{E}[\text{flips to HT}] = 4$

trap: in first-step equations, condition on the first move and add 1 for the step taken — forgetting the "1+" is the standard error.

5. Periodicity & Ergodicity

- **Period:** $d(i) = \gcd\{n \geq 1 : P^n_{ii} > 0\}$; **aperiodic** if $d(i) = 1$
- **Quick tests:** $P_{ii} > 0$ (a self-loop) $\Rightarrow$ aperiodic; walks on bipartite graphs (even cycles, chessboard moves) have period 2; a deterministic cycle of length L has period L
- **Cycle rule:** if you can return in cycle lengths a and b, period divides $\gcd(a, b)$
- **Ergodic** = irreducible + aperiodic + positive recurrent
- **Ergodic** $\Rightarrow P^n_{ij} \to \pi_j$ for *all* i, j (the chain forgets its start)
- **Periodic chains:** P^n does *not* converge, but π can still exist

trap: a stationary distribution existing does not mean P^n converges to it — convergence also needs aperiodicity.

6. Stationary Distribution $\pi = \pi P$

- **Definition:** $\pi = \pi P$, $\pi_i \geq 0$, $\sum_i \pi_i = 1$
- **How to solve:** use $n - 1$ balance equations (they are redundant) + normalization; verify with the unused equation
- **Existence/uniqueness:** irreducible and positive recurrent $\Rightarrow$ a *unique* π; no stationary distribution for null recurrent or transient chains
- **Return times:** $m_i = \mathbb{E}[\text{return time to } i] = \dfrac{1}{\pi_i}$
- **Two-state:** $\pi = \left(\dfrac{q}{p+q}, \dfrac{p}{p+q}\right)$, i.e. $\pi_1 p = \pi_2 q$
- **Detailed balance:** $\pi_i P_{ij} = \pi_j P_{ji} \Rightarrow \pi$ stationary (reversible chain)
- **Birth–death chain:** $\pi_{k+1} = \pi_k \dfrac{p_k}{q_{k+1}}$ (flow across each "doorway" balances)
- **Reflecting walk,** $p < q$**:** $\pi_k = (1 - \rho)\rho^k$, $\rho = \dfrac{p}{q}$
- **Doubly stochastic** (rows *and* columns sum to 1): π uniform
- **Random walk on a graph:** $\pi_i = \dfrac{d_i}{\sum_j d_j}$ (degree proportional)
- **Ergodic theorem:** $\dfrac{1}{n}\sum_{k=1}^{n} f(X_k) \to \sum_i \pi_i f(i)$ (long-run average = stationary average)

trap: π is a row vector satisfying $\pi = \pi P$ — solving $P\pi = \pi$ (column form) gives the wrong answer unless P is symmetric.

Random Walk Recurrence (Link to Mosteller)

- **1D symmetric walk:** recurrent, *null* recurrent (return w.p. 1, infinite mean return time)
- **2D symmetric walk:** recurrent (return probability 1) — Mosteller Problem 51
- **3D walk:** *transient*. Simple 6-neighbour lattice walk returns with probability ≈ 0.34 (Pólya); Mosteller's 8-corner version (Problem 52) ≈ 0.239
- **Biased walk ($p \neq q$) on $\mathbb{Z}$:** transient (drifts to $\pm\infty$)
- **Reflecting walk on $\{0, 1, \ldots\}$:** $p < q$ positive recurrent; $p = q$ null recurrent; $p > q$ transient

Quick Decision Guide — "Which Tool Do I Need?"

- **Given a word description** $\to$ build P (row = from), check rows sum to 1.
- **"Where am I after n steps?"** $\to \mu_0 P^n$; for one entry use row·column; for large n jump to π.
- **Specific path probability** $\to$ product of one-step probabilities (Markov property).
- **Classify first:** find communicating classes; any state with $P_{ii} = 1$ is absorbing; a class with an exit is transient; a closed finite class is positive recurrent.
- **Absorption / ruin / pattern waiting** $\to$ first-step analysis (a small linear system), not P^n.
- **Long-run fraction of time** or **mean return time** $\to$ solve $\pi = \pi P$ and use $m_i = 1/\pi_i$.
- **Chain looks like a line** $\to$ detailed balance (birth–death); **random walk on a graph** $\to$ degrees; **columns also sum to 1** $\to$ uniform.
- **Does P^n converge?** Check irreducible, aperiodic (self-loop or gcd of cycle lengths = 1), positive recurrent. If any fails, interpret π only as a long-run *time* average.
- **Infinite state space:** decide positive/null/transient from the drift *before* trying to normalize a π (a null or transient chain has no stationary distribution).

Numbers Worth Memorizing Cold

- Weather chain $\begin{pmatrix} .8 & .2 \\ .4 & .6 \end{pmatrix}$: $\pi = (\frac{2}{3}, \frac{1}{3})$, $\lambda_2 = 0.4$, $m_S = 1.5$
- Fair ruin from k of N: k/N
- Reflecting walk $p = 0.3$, $q = 0.7$: $\rho = \frac{3}{7}$, $\pi_0 = \frac{4}{7}$, $m_0 = \frac{7}{4}$
- Flips to HH $= 6$; to HT $= 4$; to HHH $= 14$
- 4-cycle walk: period 2; self-loop $\Rightarrow$ period 1
- Doubly stochastic 3×3: $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
- Graph walk: $\pi_i = d_i/(2|E|)$